



Flipperzero device retrieving data from NFC card
(Credit: <https://flipperzero.one>)

3. Transferring data. Once the connection is established, the devices can exchange data. The reader can request specific data from the tag, or the tag can send data to the reader without a request.

How secure NFC debit cards are

NFC is often used in credit and debit cards as a way to enable contactless payments. When you use an NFC-enabled credit or debit card to make a payment, the card communicates with the payment terminal using RF waves. This allows you to make a payment simply by tapping your card on the terminal, without the need to insert it into a card reader and enter a personal identification number (PIN).

NFC is based on the ISO/IEC 14443 standard, which defines the technical requirements for contactless smart cards and the communication between them and the reader. There are different types of NFC technologies that are used in credit and debit cards, including:

Type A. This is the most widely used type of NFC technology in credit and debit cards. It operates at a frequency of 13.56MHz and uses a protocol known as ISO/IEC 14443-3A.

Type B. This type of NFC technology is similar to Type A, but it uses a different protocol known as ISO/IEC 14443-3B. It is less common than Type A, but it is still used in some credit and debit cards.

FeliCa. This is a proprietary NFC technology developed by

Sony and used primarily in Japan. It also operates at a frequency of 13.56MHz but uses a protocol known as JIS X 6319-4.

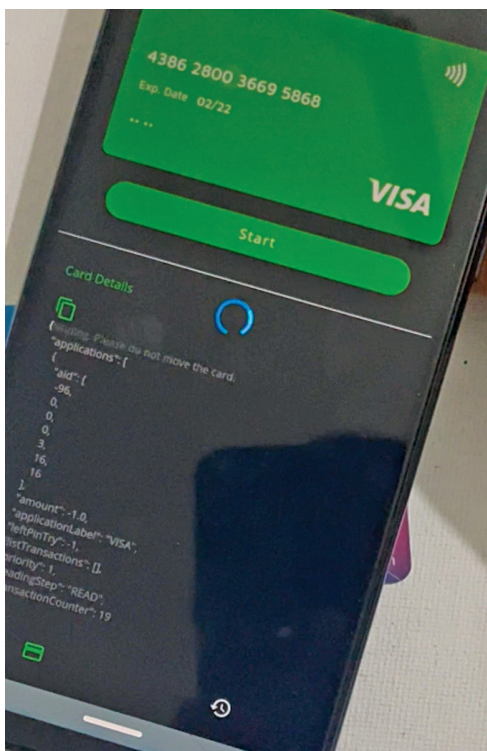
Overall, the type of NFC technology used in a credit or debit card will depend on the specific card and the payment system it is designed to work with. Regardless of the type of NFC technology used, the security measures built into NFC technology and the additional security features of credit and debit cards, such as chip-and-pin or chip-and-signature protection, help to ensure the safety and security of contactless payments.

That being said, no technology is completely foolproof, and there is always a risk that a hacker could find a way to bypass the security measures in place. To help protect yourself from potential NFC-related fraud, it is important to take steps to secure your credit and debit cards, such as keeping them in a safe location, monitoring your accounts for any suspicious activity, and reporting any lost or stolen cards as soon as possible.

Hacking NFC

There are different ways a hacker might attempt to hack NFC, such as:

Skimming. This involves using a reader and writer device to capture the data transmitted between an



The author retrieved the credit card information and transaction details using NFC on the phone and the NFC tool app

1. Initiating communication.

One of the devices, usually the one that initiates the communication, becomes an NFC reader. The other device becomes an NFC tag, which is read by the reader.

2. Establishing a connection. The reader and the tag establish a connection by exchanging a series of signals. The reader sends a request to the tag, and the tag responds by sending back its ID and other information.



User stealing data from a card using phone